

Smart Cities and the IoT: A Relationship to Build the Future

The Internet of Things (IoT) offers new opportunities for cities to use data to manage traffic, cut pollution, make better use of infrastructure and keep citizens safe.



Jitu's mother says, "Jitu, please call for an air taxi. We need to go to Connaught Place urgently for shopping." Jitu fiddles with his cellphone and the apps installed on it to book an air taxi. He then receives an alert message from Smart Waste Container Truck to deposit dustbins from his housing society. Jitu takes care of it immediately—he is a young social worker at his housing society. He downloads Clean City app from New Delhi Municipal Council's website and configures the location of his housing society in that app. Immediately, it displays the current pollution level as PM1.5 in Jitu's area.

Jitu says, "Mom, I will give a command to the smart dustbin to coordinate with the truck that will soon be here."

All this happens in New Delhi in the year 2028, when

New Delhi has become a smart city. But how can New Delhi become smart without the Internet of Things (IoT) technologies? A smart city is defined as a city that connects its physical, ICT, social and business infrastructures to leverage the collective intelligence of the city.

A city cannot become smart without the interconnection of utilities like energy, transport and ICT, waste and water supply management, connected hospitals and blood banks, connected homeland security and schools, among others.

These connections are made possible by the various layers of the IoT. There are three layers in the IoT. First is the application layer that processes, collects and displays data in various formats.